

Agile Scrum Framework

Complete guide

Introduction / Definitions

framework dedicated to project management

Scrum is a lightweight framework for generating value through adaptive solutions for complex problems.

It allows teams to focus on continuous improvement while building and iterating quickly.

Scrum workflow

Sprint Backlog

breaks large projects into smaller parts

Sprint

the team works on those parts and quickly delivers them iteratively

Sprint Review client feedback is acknowledged between iterations

product / Increment inspection

discussion with stakeholders

Product Backlog adaptation

Sprint Retrospective

inspect the way of working

identify improvements

improve effectiveness

The flexible process is supposed to avoid

It is structured around events in which the team shares its progress and problems

Its goal is to continuously deliver small parts of the project.

this strategy aims to :

- adapt the tasks to the actual requirements and obstacles
- exchange with the final client at each iteration to make sure the expectations will be met

5 Values

Focus

The team is focused on the work chosen from the backlog in order to ship deliverables by the end of the sprint

Courage

Team members must have the courage to ask questions and answer honestly in order to arrive at the best solution.

Openness

Not everything will go perfectly

Team members must be open to new ideas and opportunities that help them learn or improve the product/process

Commitment

The Scrum Team is a unit, and team members need to trust one another. Scrum team members are committed to the Sprint for its duration and are dedicated to continuous improvement of both product and process

Respect

Collaboration is the key to Scrum—and in order to support team collaboration, team members have to respect one another, the Scrum master, and the Scrum process.

6 Principles

Control over the empirical process

The team believes in transparency, inspection, and adaptation.

Self-managing

every member is empowered to take ownership of their tasks and work
shared ownership leads to more creative and dynamic teams

Collaboration

The team delivers the best results if working together during and after the Scrum sprint

Value-based prioritization

The Sprint Goal is to deliver the best business value
prioritize the work from the beginning of the process

Timeboxing

process has various time-based activities (sprint, daily standups, retrospective)

it's important to timebox work in order to move on to the next task and improve future work

Iterative development

The first version won't be perfect

by building iteratively, team will be able to adapt to customer needs and modify the product and outputs on value-based prioritization

3 Pillars

Transparency everybody knows how advanced the project is

Inspection the results/achievements are monitored on a regular basis

Adaptation the feedback determines how the project must evolve

Artifacts

An artifact is something you make, like a tool to solve a problem.

There are three artifacts in Scrum:

- Product backlog : ordered list of what is needed to improve the product
- Sprint backlog : tasks selected to be developed during the Sprint
- Product Increment / Increment : iteration of delivered product

Artifact	Commitment
Product Backlog	Product Goal
Sprint Backlog	Sprint Goal
Increment	Definition of Done

Roles / Accountabilities

- **Product Owner**

has the vision of the product

is accountable for maximizing product value

lists the requirements (product backlog)

combines the different use cases into a coherent whole

contacts the subject matter experts to refine the user stories

brings clarity about what's most important to deliver next

- **Scrum Master**

facilitate Scrum / help the Scrum Team

listen to every participant (during the daily scrums) in order to deal with the obstacles

facilitates Scrum events when needed (daily standup meetings)

hosts the sprint planning, review, and retrospective meetings.

coach → servant-leader → facilitates Scrum → helps remove impediments → helps the team improve

- **Developers**

Technical staff (developers, quality assurance, data analysts, DevOps, ...)

The Scrum Team is typically 10 or fewer people.

The staff is self-organized and collaborative

The Developers decide how to organize the work required to achieve the Sprint Goal.

Events

Sprint :

- Sprint Planning
- Daily Scrum
- Sprint Review
- Sprint Retrospective

Product backlog ⇒ **Sprint planning** ⇒ **Daily Scrum** ⇒ **Sprint review** ⇒ **Sprint retrospective**
(session) (standups)

Product backlog

List of the requirements the product has to meet

The list is defined by the product owner

The more the requirement is on top of the list, the higher priority

It also takes the dependencies into account

this list is alive, it is updated at each iteration

Sprint planning

Scrum Team joins to define the *Sprint Backlog*

Sprint Backlog is a coherent subset of the *Product Backlog*

This *Sprint Backlog* lists the requirements that the next *increment* (deliverable prototype) shall meet

Dependencies should be identified and managed because they can affect the team's ability to achieve the Sprint Goal.

For a one-month Sprint, Sprint Planning is timeboxed to a maximum of 8 hours. For shorter Sprints, the event is usually shorter.

Planning Poker - Agile estimation technique, not a mandatory Scrum event

- Each *task* is exposed on a card with its *requirements*
- For each *task*, every member of the *developers* determines secretly how much time is needed to accomplish the *task*.
- Once everyone has determined the time needed for a *task*, everybody simultaneously reveals his estimation
- The *scrum master* asks the developers with the highest and the lowest estimation to justify their choices.
- Knowing this additional information, everybody makes a new estimation
- This process is repeated until the team reaches a consensus about the time needed for this particular *task*

Once every *task* has been given a time estimate, the *Sprint Planning* is updated
It needs to be doable in 1 to 4 weeks

Sprint

fixed length of one month or less

Every morning, there is a *Daily Scrum*

At the end of the sprint, an *increment* is delivered

Daily scrum

A daily 15-minute event for Developers to inspect progress toward the Sprint Goal and adapt the Sprint Backlog.

(sometimes with the product owner)

Also called "*stand-up meeting*" because it has to be brief

- What did I do yesterday ?
- What will I do today ?
- Any impediments ?

The *scrum master* pays attention to the obstacles and finds a way to remove them

Sprint review

Once an increment is delivered, a meeting is held in which a demo is displayed to the matter experts, the client and whoever should be present

inspect the outcome of the Sprint

discuss progress toward the Product Goal

collaborate with stakeholders

determine future adaptations

The *product owner* uses the client feedback to update the *Product Backlog*

Sprint retrospective concludes the Sprint

Before starting a new *sprint* ...

... the working process is being analysed and new ideas are being expressed to provide continuous improvement

Sprint planning (2nd iteration)

Same loop goes on until the final product is being delivered

Case studies

Case Study 1 — E-commerce website

A company wants to develop an online store.

The Product Owner defines a Product Goal:

Product Goal: Create an online store allowing customers to purchase products online.

The Product Backlog contains User Stories such as:

- As a customer, I want to browse products by category.
- As a customer, I want to add products to my shopping cart.
- As a customer, I want to pay online.
- As a customer, I want to receive an order confirmation.

During Sprint Planning, the Developers select a subset of these items and define a Sprint Goal:

Sprint Goal: Enable customers to browse products and add them to a shopping cart.

At the end of the Sprint, the team presents the Increment during the Sprint Review.

Stakeholder feedback may lead to adaptations of the Product Backlog.

During the Sprint Retrospective, the Scrum Team identifies improvements to its way of working.

Case Study 2 — Machine Learning project

A company wants to predict customer churn.

The Product Goal is:

Product Goal: Provide a reliable system for identifying customers at risk of leaving.

The Product Backlog may contain items such as:

- collect and clean historical customer data;
- analyse missing values and data quality;
- create relevant features;
- train a baseline model;
- evaluate model performance;
- expose predictions through an API;
- monitor model performance.

A Sprint Goal could be:

Sprint Goal: Build and evaluate a baseline churn prediction model.

The Increment could contain:

- a cleaned dataset;
- a reproducible preprocessing pipeline;
- a trained baseline model;
- evaluation metrics;
- technical documentation.

During the Sprint Review, stakeholders inspect the results and discuss whether the model provides sufficient value.

The Product Backlog can then be adapted according to the findings.

Case Study 3 — Changing requirements

A development team is building a mobile application.

During a Sprint, a stakeholder discovers that users need an additional authentication method.

The team does not simply add the new requirement to the Sprint without considering the Sprint Goal.

Instead, the new requirement is added to the Product Backlog and considered during future Product Backlog refinement and Sprint Planning.

If the Sprint Goal remains valid, the Developers continue working toward it.

If the Sprint Goal becomes obsolete, the Product Owner may cancel the Sprint.

This illustrates an important Scrum principle:

The Product Backlog can change, while the Sprint provides a stable goal for the Developers.

Key words / Common Scrum practices

user story

A short description of a requirement written from the user's perspective.

A common format is:

As a [type of user], I want [goal] so that [benefit].

User stories describe **what the user needs and why**, rather than how the feature should be implemented.

Example:

As a customer, I want to receive an email confirmation after placing an order so that I know my order has been successfully registered.

A User Story is not necessarily a mandatory Scrum artifact or a specific format required by Scrum. It is a common technique for expressing Product Backlog Items.

story points

Story Points are a relative measure used to estimate the overall effort required to complete a Product Backlog Item.

They take into account factors such as:

- complexity
- amount of work
- uncertainty
- technical difficulty

Story Points do **not** represent a fixed amount of time.

Teams often use a sequence such as:

1, 2, 3, 5, 8, 13, 21

The values are relative: an item estimated at 8 Story Points is considered more demanding than one estimated at 3 Story Points.

Story Points are a common estimation technique, but Scrum does not prescribe their use.

Burndown chart

A Burndown Chart is a visual tool used to track the amount of work remaining over time.

- X-axis: time
- Y-axis: remaining work

The chart generally shows a downward trend as work is completed.

Example:

40 → 32 → 25 → 18 → 10 → 0 Story Points

A Burndown Chart can help the team visualize progress and identify whether the planned work is likely to be completed within the Sprint.

However, a Burndown Chart is not a mandatory Scrum artifact. It is a common project management and Agile visualization technique.

Kanban method

The [Kanban framework](#) is a subset of Agile, **not a part of Scrum**

It helps the team visualize work moving through different stages until completion

DoR - Definition of Ready - **not mandatory in official Scrum Guide 2020**

When a requirement is considered sufficiently defined to be shifted from the Product Backlog to the Sprint Backlog

It must be clear, precisely described, with no remaining ambiguity, illustrated by examples
So that the development and testing teams understand what is wanted

DoD - Definition of Done

- When a requirement is considered done
- developed
- tested : all unit tests, integration tests and system tests are OK
- documented
- presented via demo

Determines whether work is part of a usable Increment.